

AI/ML Internship – Day 64 Notes & Tasks

Module 8: Building an AI Chatbot using the OpenAI API

Part 1: Recap of Day 63

Yesterday we learned:

-  AI Chatbots
-  Intent Classification
-  Training Data
-  Feature Extraction
-  Machine Learning Concepts

Today we will learn:

- What is an API?
 - What is the OpenAI API?
 - How ChatGPT works
 - Building an AI chatbot using Python
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Part 2: What is an API?

Definition

API (Application Programming Interface) is a bridge that allows two software applications to communicate with each other.

Simple Definition

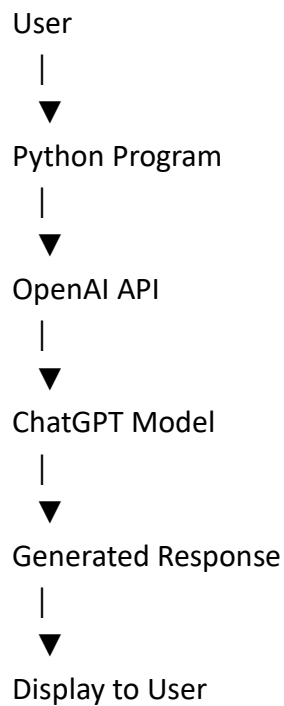
An API is like a **waiter in a restaurant**.

- You (User) → Place an order
- Waiter (API) → Takes the order to the kitchen
- Kitchen (Server/AI Model) → Prepares the food
- Waiter (API) → Brings the food back to you

Similarly:

- User → Sends a message
 - API → Sends it to ChatGPT
 - ChatGPT → Generates a response
 - API → Returns the response
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Part 3: API Workflow



Part 4: What is the OpenAI API?

The OpenAI API allows developers to integrate AI models into their own applications.

Using the API, developers can build:

- AI Chatbots
 - Virtual Assistants
 - AI Tutors
 - Content Generators
 - Code Assistants
 - Customer Support Systems
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Part 5: Real-World Applications

Companies use AI APIs for:

- Customer Support
 - Email Writing
 - Code Generation
 - Document Summarization
 - Translation
 - Education
 - Healthcare Assistance
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Part 6: Prerequisites

Before using the OpenAI API, you need:

- Python installed
 - Internet connection
 - OpenAI API Key
 - OpenAI Python SDK
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Part 7: Install the OpenAI Library

`pip install openai`

Part 8: API Key

An **API Key** is a secret code that authenticates your application.

Example:

`sk-xxxxxxxxxxxxxxxxxxxxxxxxxx`

 Never share your API key publicly.

Part 9: Basic Python Example

```
from openai import OpenAI

client = OpenAI(api_key="YOUR_API_KEY")

response = client.responses.create(
    model="gpt-4.1-mini",
    input="Hello! Who are you?"
)

print(response.output_text)
```

Part 10: Building a Simple Chatbot

```
from openai import OpenAI

client = OpenAI(api_key="YOUR_API_KEY")

print("🤖 AI Chatbot")
print("Type 'bye' to exit.\n")

while True:

    user = input("You: ")

    if user.lower() == "bye":
        print("Bot: Goodbye!")
        break

    response = client.responses.create(
        model="gpt-4.1-mini",
        input=user
    )

    print("Bot:", response.output_text)
```

Part 11: Example Conversation

You: What is Artificial Intelligence?

Bot:

Artificial Intelligence is the simulation of human intelligence by machines that can learn, reason, and solve problems.

You:

Who invented Python?

Bot:

Python was created by Guido van Rossum in 1991.

You:

Explain Machine Learning.

Bot:

Machine Learning is a branch of AI where computers learn patterns from data and make predictions without being explicitly programmed.

Part 12: Why is this Better than Rule-Based Chatbots?

Rule-Based Chatbot

Only answers predefined questions.

AI Chatbot

Can answer millions of different questions.

Rule-Based

Uses if-else

AI Chatbot

Uses Large Language Models

Rule-Based

No learning

AI Chatbot

Trained on massive datasets

📌 Part 13: Advantages

- ✓ Understands natural language
 - ✓ Handles long conversations
 - ✓ Can explain concepts
 - ✓ Can write code
 - ✓ Can summarize documents
 - ✓ Can translate languages
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📌 Part 14: Limitations

- ✗ Requires an internet connection
 - ✗ API usage may have costs
 - ✗ Responses can occasionally be inaccurate
 - ✗ Sensitive information should not be shared
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📌 Part 15: Best Practices

- Keep your API key private.
 - Validate user input before sending requests.
 - Handle errors such as network failures.
 - Do not hardcode API keys in production applications.
 - Store API keys in environment variables.
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📌 Part 16: Difference Between Traditional Chatbots and LLM Chatbots

Traditional Chatbot LLM Chatbot

Fixed rules Understands natural language

Traditional Chatbot LLM Chatbot

Limited responses	Flexible responses
Uses keywords	Uses Large Language Models
Simple logic	Context-aware conversations
Best for FAQs	Best for complex conversations

Part 17: Theory Questions

What is an API?

An API is a communication interface that allows different software applications to exchange data and services.

What is the OpenAI API?

The OpenAI API allows developers to use AI models in their own applications.

Why is an API key required?

An API key authenticates your application and authorizes access to the AI service.

Name three applications of the OpenAI API.

- AI Chatbots
 - Code Assistants
 - Content Generation
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Why are LLM-based chatbots better than rule-based chatbots?

They understand natural language, generate flexible responses, and can answer a much wider variety of questions.

Part 18: Interview Questions

What is an API?

An API is a set of rules that enables communication between software applications.

What is an API Key?

An API key is a unique authentication token used to access an API securely.

What is the difference between a Rule-Based chatbot and an LLM chatbot?

A rule-based chatbot follows predefined rules, while an LLM chatbot generates responses using a trained language model.

Why should API keys be kept secret?

If someone obtains your API key, they can misuse your account and generate unwanted charges.

What are Large Language Models?

Large Language Models (LLMs) are AI models trained on massive text datasets to understand and generate human-like language.

Tasks for Day 64

Theory

1. Define API.
 2. Explain the OpenAI API.
 3. What is an API key?
 4. Explain the workflow of an AI chatbot using an API.
 5. Compare Rule-Based and LLM-based chatbots.
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Practical Task 1

Install the OpenAI Python library.

```
pip install openai
```

Practical Task 2

Write a Python program that sends a greeting to the AI model and displays the response.

Practical Task 3

Modify the chatbot to continue the conversation until the user types **bye**.

Practical Task 4

Create a chatbot that answers questions related to:

- Artificial Intelligence
 - Python
 - Machine Learning
 - Data Science
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Practical Task 5

Draw the API communication workflow.